

Customer Churn Prediction

Retail churn prediction project. Dataset has ~50k customer records with purchase history, demographics, and a churn label.

What I did

Started with some EDA to understand the data, then built features (RFM indicators mostly), handled class imbalance with SMOTE, and trained logistic regression + random forest models. Random forest ended up being significantly better.

Final results:

- Accuracy: 87%
- AUC-ROC: 0.91
- Used k-fold CV and GridSearchCV for tuning

Visualized the churn risk segments in Tableau — the dashboard made it easier to communicate findings to non-technical stakeholders.

Files

- `01_eda_and_cleaning.py` — initial exploration, null checks, distributions
- `02_feature_engineering.py` — RFM features, encoding, SMOTE
- `03_modeling.py` — model training, evaluation, hyperparameter tuning
- `churn_sql_queries.sql` — RFM scoring and cohort queries in SQL
- `requirements.txt` — dependencies

Requirements

```
pandas
numpy
scikit-learn
imbalanced-learn
matplotlib
```

Notes

The dataset is simulated retail data. Class imbalance was around 78/22 so SMOTE helped noticeably — without it the model was just predicting majority class most of the time.